

Experiment 5. Affine Caesar Cipher

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int gcd(int a, int b)
```

```
{
```

```
    while(b != 0)
```

```
    {
```

```
        int t = b;
```

```
        b = a % b;
```

```
        a = t;
```

```
    }
```

```
    return a;
```

```
}
```

```
int main()
```

```
{
```

```
    char text[100];
```

```
    int a, b, i, p, c;
```

```
    printf("Enter plaintext: ");
```

```
    scanf("%s", text);
```

```
    printf("Enter a: ");
```

```
    scanf("%d", &a);
```

```
    printf("Enter b: ");
```

```
scanf("%d", &b);
```

```
if(gcd(a, 26) != 1)
```

```
{
```

```
    printf("Invalid value of a");
```

```
    return 0;
```

```
}
```

```
for(i = 0; text[i] != '\0'; i++)
```

```
{
```

```
    p = text[i] - 'A';
```

```
    c = (a * p + b) % 26;
```

```
    text[i] = c + 'A';
```

```
}
```

```
printf("Ciphertext: %s", text);
```

```
return 0;
```

```
}
```

